

Lab 3 - Routing Protocol

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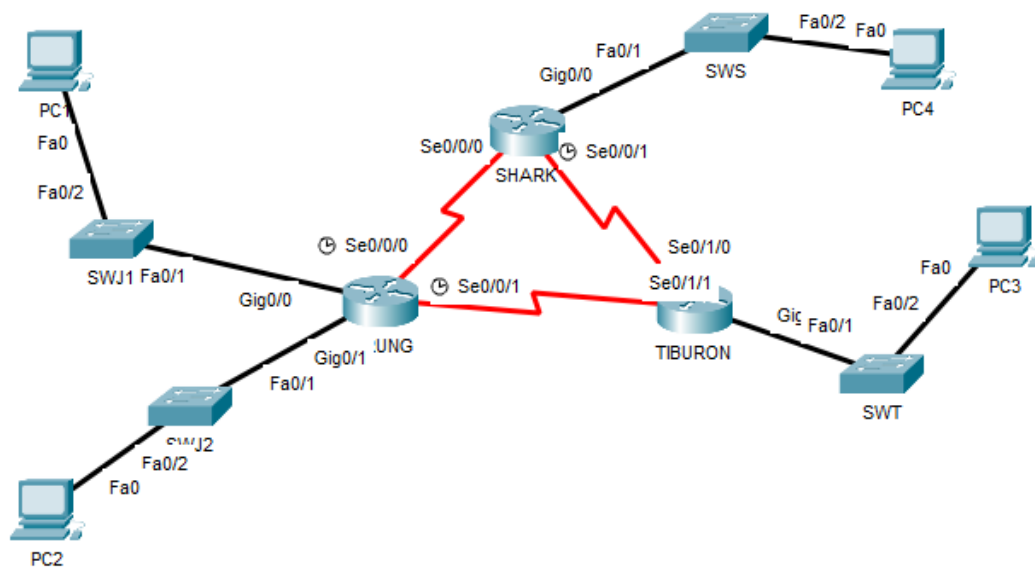
MATRIC NUM: A23CS0201

SECTION: 02

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	
2		Se0/0/1	172.18.110.197	255.255.255.252	
3		G0/0	172.18.110.1	255.255.255.224	
4		G0/1	172.18.110.33	255.255.255.224	
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	
6		Se0/1/0	172.18.110.202	255.255.255.252	
7		G0/0	172.18.110.129	255.255.255.192	
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	
9		Se0/0/1	172.18.110.201	255.255.255.252	
10		G0/0	172.18.110.65	255.255.255.192	
11	PC1	-	172.18.110.2	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.34	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.130	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.66	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

1. Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
2. Using the IP addresses calculated, configure the devices with the appropriate information.

JERUNG

1. LAN 1 (20 Host)

$$\text{Subnet} : 20 \approx 32 = 2^5 \\ = 5 \text{ bits}$$

$$\text{subnet} = 32 - 5 = 127$$

→ 1010 1100 . 0001 0010 . 0110 1110 . 000 00000

network portion

host portion

* subnet mask

$$= 1111 1111 . 1111 1111 . 1111 1111 . 111 00000 \\ (255 . 255 . 255 . 224)$$

* network address

$$= 1010 1100 . 0001 0010 . 0110 1110 . 000 00000 \\ (172 . 18 . 110 . 0 / 27)$$

* Broadcast address

$$= 1010 1100 . 0001 0010 . 0110 1110 . 000 11111 \\ (172 . 18 . 110 . 31 / 27)$$

* usable address

$$172 . 18 . 110 . 1 \text{ — } 172 . 18 . 110 . 30$$

[60/0]

2. LAN 2 (20 host)

$$\text{Subnet} : 30 \approx 32 = 2^5 \\ = 5 \text{ bits}$$

$$\text{Subnet} = 32 - 5 = /27$$

$\rightarrow 1010 \ 1100 . 0001 \ 0010 . 0110 \ 1110 . 001$ network
portion | host
portion
 00000

* Subnet mask

$$= 1111 \ 1111 . 1111 \ 1111 . 1111 \ 1111 . 1111 \ 000000 \\ (255 . 255 . 255 . 224)$$

* network address

$$= 1010 \ 1100 . 0001 \ 0010 . 0110 \ 1110 . 001 \ 000000 \\ (172 . 18 . 110 . 32 / 27)$$

* Broadcast address

$$= 1010 \ 1100 . 0001 \ 0010 . 0110 \ 1110 . 001 \ 111111 \\ (172 . 18 . 110 . 63 / 27)$$

* usable address

$$172 . 18 . 110 . 33 \text{ --- } 172 . 18 . 110 . 62$$

[60 / 1]

SHARK

1. LAN (60 Hosts)

$$\text{subnet} : 60 \approx 64 = 2^6$$

= 6 bits

$$\text{subnet} = 32 - 6 = /26$$

$$\rightarrow 1010 \ 1100 . 0001 \ 0010 . 0110 \quad \begin{array}{c} \text{network} \\ \text{portion} \end{array} \quad 110 . 01 \quad \begin{array}{c} \text{host} \\ \text{portion} \end{array} \quad 000000$$

* Subnet mask

$$= 1111 \ 1111 . 1111 \ 1111 . 1111 \ 1111 . 11 \ 000000$$

$$(255 . 255 . 255 . 192)$$

* network address

$$= 1010 \ 1100 . 0001 \ 0010 . 0110 \ 1110 . 01 \ 000000$$

$$(172 . 18 . 110 . 64 /26)$$

* Broadcast address

$$= 1010 \ 1100 . 0001 \ 0010 . 0110 \ 1110 . 01 \ 000000$$

$$(172 . 18 . 110 . 127 /26)$$

* usable address

$$172 . 18 . 110 . 65 - 172 . 18 . 110 . 126$$

$$[60/0]$$

TI BURON

1. LAN (50 host)

$$\text{subnet} : 50 \approx 64 = 2^6$$

$$= 6 \text{ bits}$$

$$\text{subnet} = 32 - 6 = /26$$

$$\rightarrow 1010 \ 1100 \cdot 0001 \ 0010 \cdot 0110 \ 1110 \cdot 10 \quad \begin{array}{c} \text{network} \\ \text{portion} \end{array} \bigg| \begin{array}{c} \text{host} \\ \text{portion} \end{array} 000000$$

* subnet mask

$$= 1111 \ 1111 \cdot 1111 \ 1111 \cdot 1111 \ 1111 \cdot 11 \ 000000$$

$$(255 \cdot 255 \cdot 255 \cdot 128/26)$$

* Network address

$$= 1010 \ 1100 \cdot 0001 \ 0010 \cdot 0110 \ 1110 \cdot 10 \ 000000$$

$$(172 \cdot 18 \cdot 110 \cdot 128 / 26)$$

* Broadcast address

$$= 1010 \ 1100 \cdot 0001 \ 0010 \cdot 0110 \ 1110 \cdot 10 \ 111111$$

$$(172 \cdot 18 \cdot 110 \cdot 191 / 26)$$

* usable address

$$172 \cdot 18 \cdot 110 \cdot 129 - 172 \cdot 18 \cdot 110 \cdot 190$$

$$[60/0]$$

JERUNG - SHARK

2 Host

$$\text{subnet} : 2^1 = 2 \\ = 2 \text{ bits}$$

$$\text{subnet} = 32 - 2 = 180$$

$$\rightarrow 1010 \ 1100 \cdot 0001 \ 0010 \cdot 0110 \ 1110 \cdot 11 \ 0000 \ 00$$

network position	host position
1010 1100 · 0001 0010 · 0110 1110 · 11 0000 00	00

* subnet mask

$$= 1111 \ 1111 \cdot 1111 \ 1111 \cdot 1111 \ 1111 \cdot 1111 \ 1100 \\ (255 \cdot 255 \cdot 255 \cdot 252)$$

* Network address

$$= 1010 \ 1100 \cdot 0001 \ 0010 \cdot 0010 \ 1110 \cdot 11 \ 0000 \ 00 \\ (172 \cdot 18 \cdot 110 \cdot 192 / 30)$$

* Broadcast address

$$= 1010 \ 1100 \cdot 0001 \ 0010 \cdot 0110 \ 1110 \cdot 110000 \ 11 \\ (172 \cdot 18 \cdot 110 \cdot 195 / 30)$$

* Usable address

$$= 172 \cdot 18 \cdot 110 \cdot 193 \quad - \quad 172 \cdot 18 \cdot 110 \cdot 194 \\ [\text{SE } 0/0/0 \text{ MASK}]$$

JERUNG - TIBURON

2 Host

$$\text{subnet} : 2^1 = 2 \\ = 2 \text{ bits}$$

$$\text{subnet} : 32 - 2 = /30$$

	network portion	host portion
→ 1010 1100 . 0001 0010 . 0110 1110 . 110001		00

* subnet mask

$$= 1111 1111 . 1111 1111 . 1111 1111 . 11111100 \\ (255.255.255.252)$$

* network address

$$= 1010 1100 . 0001 0010 . 0110 1110 . 11000100 \\ (172.18.110.196 /30)$$

* Broadcast Address

$$= 1010 1100 . 0001 0010 . 0110 1110 . 11000111 \\ (172.18.110.199 /30)$$

* usable address

$$172.18.110.197 - 172.18.110.198 \\ [SEO / 1 / 1 \text{ TIBURON}]$$

SHARK - TIBURON

2 host

$$\text{subnet} : 2^1 = 2 \\ = 2 \text{ bits}$$

$$\text{subnet} = 32 - 2 = 130$$

→ 1010 1100 . 0001 0010 . 0110 1110 . 11 0010

network portion	host portion
1010 1100 . 0001 0010 . 0110 1110 . 11 0010	00

* subnet mask

$$= 1111 1111 . 1111 1111 . 1111 1111 . 1111 00 \\ (255 . 255 . 255 . 252 / 30)$$

* network address

$$= 1010 1100 . 0001 0010 . 0110 1110 . 11 0010 00 \\ (172 . 18 . 110 . 200 / 30)$$

* broadcast address

$$= 1010 1100 . 0001 0010 . 0110 1110 . 11 0010 11 \\ (172 . 18 . 110 . 203 / 30)$$

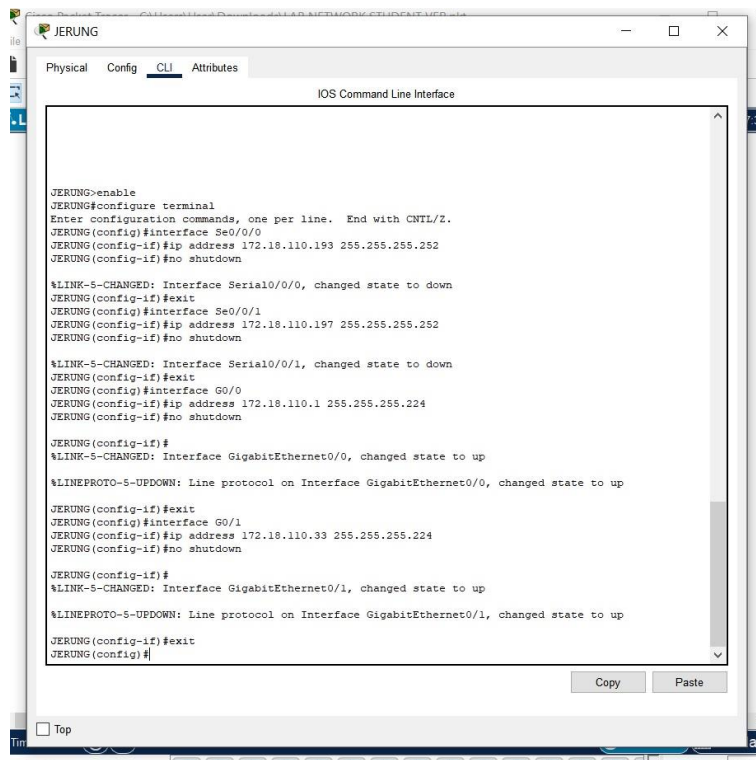
* usable address

$$172 . 18 . 110 . 201 - 172 . 18 . 110 . 202$$

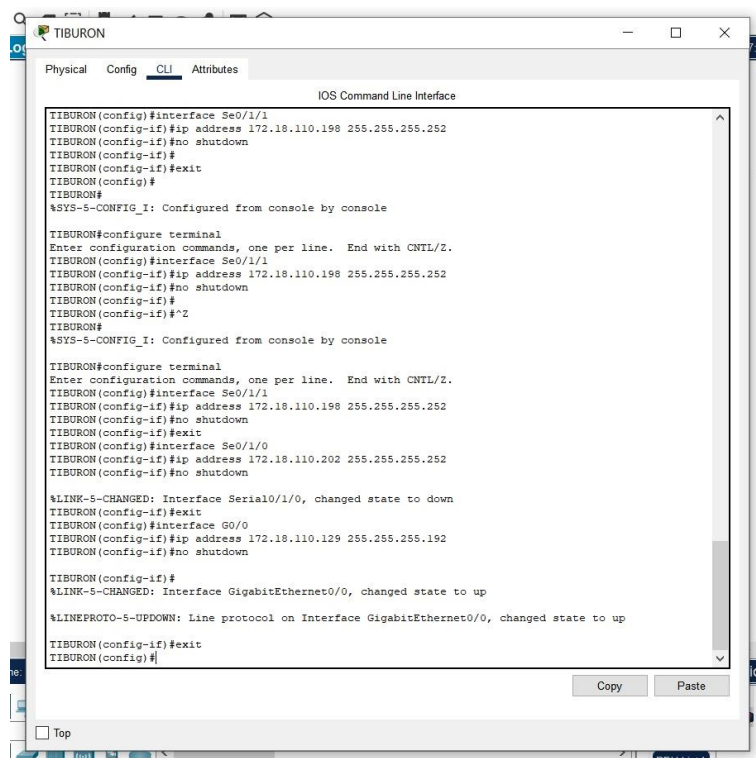
[SED / 1 / 0 TIBURON]

HKMS12425

JERUNG

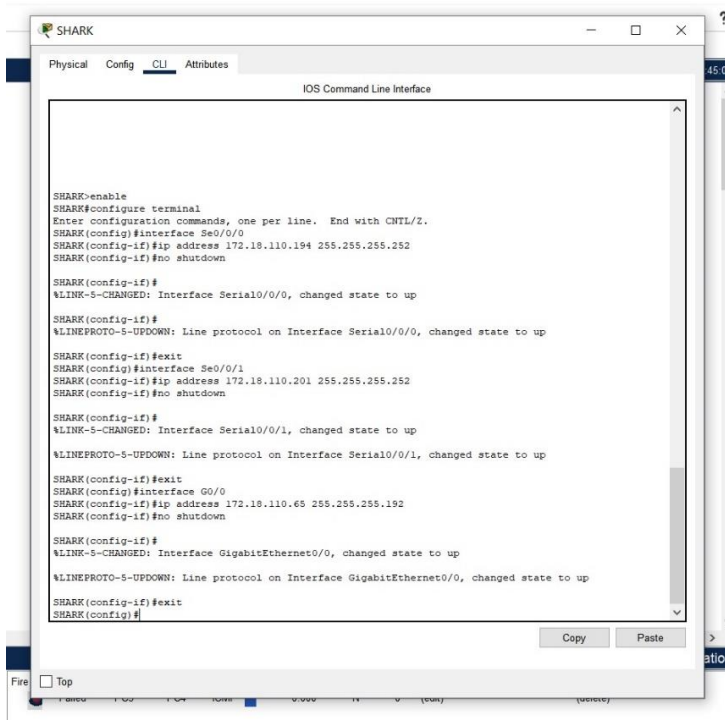


TIBURON

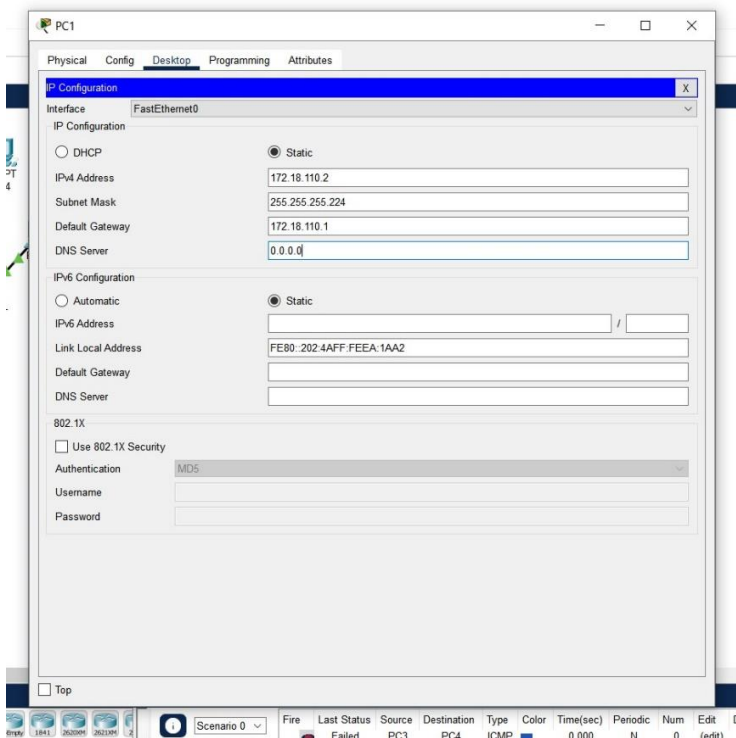


HKMS12425

SHARK



PC1



HKMS12425

PC2

The screenshot shows the 'PC2' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing settings for the 'FastEthernet0' interface. The 'Static' radio button is selected under 'IP Configuration'. The IPv4 Address is 172.18.110.34, Subnet Mask is 255.255.255.224, Default Gateway is 172.18.110.33, and DNS Server is 0.0.0.0. The 'IPv6 Configuration' section also has 'Static' selected, with an empty IPv6 Address field, Link Local Address FE80::2E0:8FFF:FED8:CD1, and empty Default Gateway and DNS Server fields. The '802.1X' section has 'Use 802.1X Security' unchecked, Authentication set to 'MD5', and empty Username and Password fields. A 'Top' button is at the bottom left.

Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	172.18.110.34
Subnet Mask	255.255.255.224
Default Gateway	172.18.110.33
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::2E0:8FFF:FED8:CD1
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

☐ Top

PC3

The screenshot shows the 'PC3' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing settings for the 'FastEthernet0' interface. The 'Static' radio button is selected under 'IP Configuration'. The IPv4 Address is 172.18.110.130, Subnet Mask is 255.255.255.192, Default Gateway is 172.18.110.129, and DNS Server is 0.0.0.0. The 'IPv6 Configuration' section also has 'Static' selected, with an empty IPv6 Address field, Link Local Address FE80::204:9AFF:FE85:832, and empty Default Gateway and DNS Server fields. The '802.1X' section has 'Use 802.1X Security' unchecked, Authentication set to 'MD5', and empty Username and Password fields. A 'Top' button is at the bottom left.

Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	172.18.110.130
Subnet Mask	255.255.255.192
Default Gateway	172.18.110.129
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::204:9AFF:FE85:832
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

☐ Top

HKMS12425

PC4

The screenshot shows a configuration window for a device named PC4. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Config tab is active, and the IP Configuration section is expanded. The interface selected is FastEthernet0. The IP Configuration section has two radio buttons: DHCP and Static. The Static option is selected. The fields for Static IP configuration are filled with the following values: IPv4 Address: 172.18.110.66, Subnet Mask: 255.255.255.192, Default Gateway: 172.18.110.65, and DNS Server: 0.0.0.0. The IPv6 Configuration section also has two radio buttons: Automatic and Static. The Static option is selected. The fields for Static IPv6 configuration are filled with the following values: IPv6 Address: (empty), Link Local Address: FE80::200:CFF:FEAA:16C8, Default Gateway: (empty), and DNS Server: (empty). The 802.1X section has a checkbox for Use 802.1X Security, which is unchecked. The Authentication dropdown menu is set to MD5. The Username and Password fields are empty. A Top button is located at the bottom left of the window.

PC4

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 172.18.110.66

Subnet Mask: 255.255.255.192

Default Gateway: 172.18.110.65

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::200:CFF:FEAA:16C8

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
 - a. CLI
 - i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```

SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0

```

Figure A

JERUNG

```

JERUNG>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.33/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

```

TIBURON

```

TIBURON>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.128/26 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0

```

SHARK

```

SHARK>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

```


b. PT tool

- i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

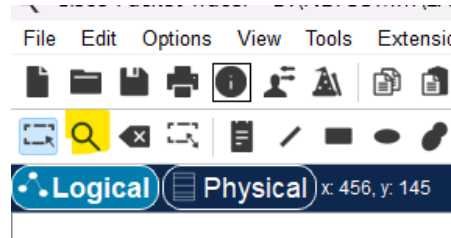


Figure B

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

PT TOOL

JERUNG

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

HKMS12425

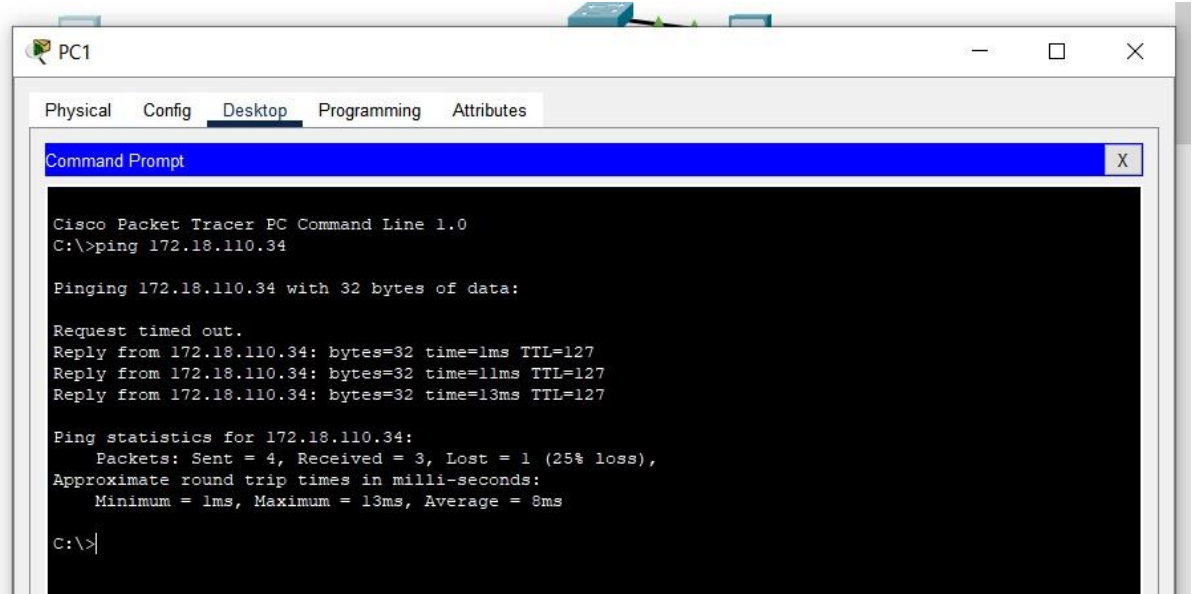
TIBURON

Routing Table for TIBURON					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.196/30	Serial0/1/1	---	0/0	
L	172.18.110.198/32	Serial0/1/1	---	0/0	
C	172.18.110.200/30	Serial0/1/0	---	0/0	
L	172.18.110.202/32	Serial0/1/0	---	0/0	

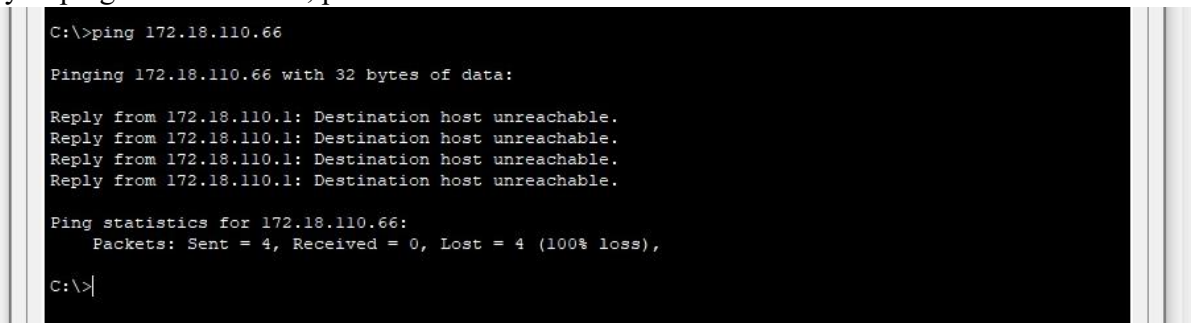
SHARK

Routing Table for SHARK					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.194/32	Serial0/0/0	---	0/0	
C	172.18.110.200/30	Serial0/0/1	---	0/0	
L	172.18.110.201/32	Serial0/0/1	---	0/0	

2. Try to ping PC2 from PC1, paste the results here.



3. Try to ping PC4 from PC1, paste the results here.



4. Explain the reason(s) behind the results.

The observed results occur because PCs on different subnets cannot communicate directly without appropriate routing configurations. When PC1 pings PC2, the packets are successfully routed because the routers have routing information for PC2's network. However, when PC1 tries to ping PC4, the packets are dropped because the routers lack routing information for the destination network. Routers only forward packets to networks listed in their routing tables. If a router cannot find a route to the destination network, it discards the packet and sends an "unreachable" message.

5. What needs to be done to ensure all PCs can ping each other successfully?

Routing Information Protocol (RIP) should be enabled on all routers.

Task 2 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.164/32	GigabitEthernet0/0	---	0/0

Figure D

- In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- What can you say about the addresses used in the 'network' instructions in Figure E?

It indicates that the router JERUNG is advertising the following subnets:

172.18.110.128/27 (172.18.110.128 -> 172.18.110.159)
 172.18.110.160/27 (172.18.110.160 -> 172.18.110.191)
 172.18.110.192/27 (172.18.110.192 -> 172.18.110.223)
 172.18.110.196/27 (172.18.110.192 -> 172.18.110.223)

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

- a. What are the changes seen in TIBURON?
3 type R (RIP) is added
- b. What are the Networks with type R in TIBURON and JERUNG?
- JERUNG
172.18.110.128/26
172.18.110.200/30
- TIBURON
172.18.110.0/27
172.18.110.32/27
172.18.110.192/30

- c. Ping PC3 from PC1. Was it successful?

Yes

```
C:\>ping 172.18.110.130

Pinging 172.18.110.130 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.130: bytes=32 time=1ms TTL=126
Reply from 172.18.110.130: bytes=32 time=1ms TTL=126
Reply from 172.18.110.130: bytes=32 time=15ms TTL=126

Ping statistics for 172.18.110.130:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 15ms, Average = 5ms
```

- d. Ping PC4 from PC1. Was it successful?

No

```
C:\>ping 172.18.110.66

Pinging 172.18.110.66 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.

Ping statistics for 172.18.110.66:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- e. Explain the reasons for your answer.

The first packet fails while pinging PC3 from PC1, but the next packets are successful. This is probably because updating the routing table or clearing the ARP cache takes time. It could take a while for the router to process the forwarding and routing. The packet to its assigned correct location. The destination host's MAC address is resolved during the initial ping, which causes a brief delay and drops the first packet. Due to improper RIP configuration on all routers, all packets sent from PC1 to PC4 are lost. The SHARK router does not notify the other routers of the networks it is connected to since it is not set up with RIP. Therefore, packets from PC1 fail to get to PC4.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK>enable
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#network 172.18.110.64
SHARK(config-router)#no auto-summary
SHARK(config-router)#
```

- g. Open router SHARK's routing table and paste here.

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

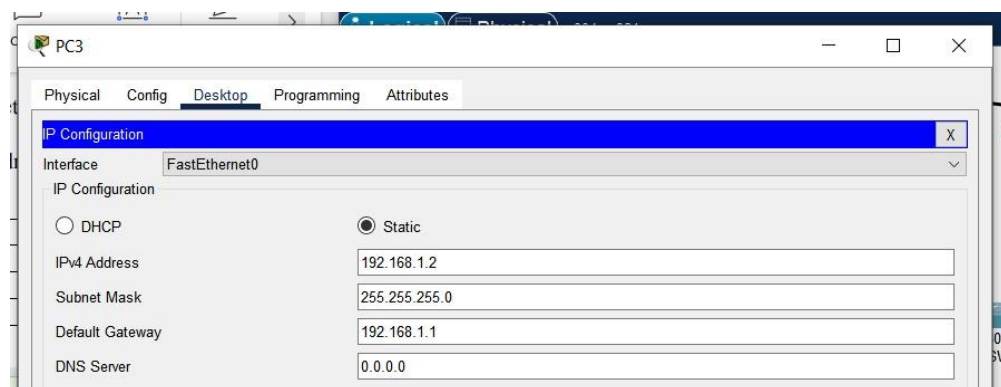
PC	Ping result
1	<pre> Cisco Packet Tracer PC Command Line 1.0 C:\>ping 172.18.110.2 Pinging 172.18.110.2 with 32 bytes of data: Reply from 172.18.110.2: bytes=32 time=1ms TTL=126 Reply from 172.18.110.2: bytes=32 time=1ms TTL=126 Reply from 172.18.110.2: bytes=32 time=2ms TTL=126 Reply from 172.18.110.2: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 2ms, Average = 1ms C:\>ping 172.18.110.2 Pinging 172.18.110.2 with 32 bytes of data: Reply from 172.18.110.2: bytes=32 time=17ms TTL=126 Reply from 172.18.110.2: bytes=32 time=1ms TTL=126 Reply from 172.18.110.2: bytes=32 time=1ms TTL=126 Reply from 172.18.110.2: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 17ms, Average = 5ms C:\> </pre>

2	<pre> C:\>ping 172.18.110.34 Pinging 172.18.110.34 with 32 bytes of data: Request timed out. Reply from 172.18.110.34: bytes=32 time=10ms TTL=126 Reply from 172.18.110.34: bytes=32 time=1ms TTL=126 Reply from 172.18.110.34: bytes=32 time=14ms TTL=126 Ping statistics for 172.18.110.34: Packets: Sent = 4, Received = 3, Lost = 1 (25% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 14ms, Average = 8ms C:\>ping 172.18.110.34 Pinging 172.18.110.34 with 32 bytes of data: Reply from 172.18.110.34: bytes=32 time=25ms TTL=126 Reply from 172.18.110.34: bytes=32 time=1ms TTL=126 Reply from 172.18.110.34: bytes=32 time=1ms TTL=126 Reply from 172.18.110.34: bytes=32 time=31ms TTL=126 Ping statistics for 172.18.110.34: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 31ms, Average = 14ms </pre>
3	<pre> C:\>ping 172.18.110.130 Pinging 172.18.110.130 with 32 bytes of data: Reply from 172.18.110.130: bytes=32 time=1ms TTL=126 Reply from 172.18.110.130: bytes=32 time=29ms TTL=126 Reply from 172.18.110.130: bytes=32 time=1ms TTL=126 Reply from 172.18.110.130: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.130: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 29ms, Average = 8ms C:\>ping 172.18.110.130 Pinging 172.18.110.130 with 32 bytes of data: Reply from 172.18.110.130: bytes=32 time=23ms TTL=126 Reply from 172.18.110.130: bytes=32 time=12ms TTL=126 Reply from 172.18.110.130: bytes=32 time=1ms TTL=126 Reply from 172.18.110.130: bytes=32 time=15ms TTL=126 Ping statistics for 172.18.110.130: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 23ms, Average = 12ms </pre>

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: 192.168.1.0/24
 - b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.2
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1



- c. After this change, can PC4 and PC1 ping PC3?
 Not possible. This is because SHARK and JERUNG don't know anything about the new network after TIBURON's G0/0 interface's IP address was changed and a new subnet was assigned. Without updated routing information, packets intended for PC3 cannot be forwarded correctly, leading to failed pings.

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

SHARK

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/16
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

TIBURON

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

- e. Referring to the routing table, explain your findings.
 A new subnet, 192.168.1.0/25, is being assigned to TIBURON's G0/0 interface. The other routers do not know the path to the new network since this subnet is not part of their current RIP configuration. Packets destined for 192.168.1.0/25 are dropped.

- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

Update RIP configuration for all routers.

- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON>enable
TIBURON#configure-terminal
Translating "configure-terminal"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#interface G0/0
TIBURON(config-if)#ip address 192.168.1.1 255.255.255.0
TIBURON(config-if)#
TIBURON(config-if)#exit
TIBURON(config)#
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#
```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	<pre>Cisco Packet Tracer PC Command Line 1.0 C:\>ping 172.18.110.2 Pinging 172.18.110.2 with 32 bytes of data: Reply from 172.18.110.2: bytes=32 time=1ms TTL=126 Reply from 172.18.110.2: bytes=32 time=14ms TTL=126 Reply from 172.18.110.2: bytes=32 time=1ms TTL=126 Reply from 172.18.110.2: bytes=32 time=2ms TTL=126 Ping statistics for 172.18.110.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 14ms, Average = 4ms C:\>ping 172.18.110.2 Pinging 172.18.110.2 with 32 bytes of data: Reply from 172.18.110.2: bytes=32 time=18ms TTL=126 Reply from 172.18.110.2: bytes=32 time=12ms TTL=126 Reply from 172.18.110.2: bytes=32 time=15ms TTL=126 Reply from 172.18.110.2: bytes=32 time=14ms TTL=126 Ping statistics for 172.18.110.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 12ms, Maximum = 18ms, Average = 14ms</pre>

2	<pre> C:\>ping 172.18.110.34 Pinging 172.18.110.34 with 32 bytes of data: Reply from 172.18.110.34: bytes=32 time=21ms TTL=126 Reply from 172.18.110.34: bytes=32 time=1ms TTL=126 Reply from 172.18.110.34: bytes=32 time=10ms TTL=126 Reply from 172.18.110.34: bytes=32 time=14ms TTL=126 Ping statistics for 172.18.110.34: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 21ms, Average = 11ms C:\>ping 172.18.110.34 Pinging 172.18.110.34 with 32 bytes of data: Reply from 172.18.110.34: bytes=32 time=23ms TTL=126 Reply from 172.18.110.34: bytes=32 time=11ms TTL=126 Reply from 172.18.110.34: bytes=32 time=28ms TTL=126 Reply from 172.18.110.34: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.34: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 28ms, Average = 15ms </pre>
4	<pre> C:\>ping 172.18.110.66 Pinging 172.18.110.66 with 32 bytes of data: Reply from 172.18.110.66: bytes=32 time=31ms TTL=126 Reply from 172.18.110.66: bytes=32 time=22ms TTL=126 Reply from 172.18.110.66: bytes=32 time=1ms TTL=126 Reply from 172.18.110.66: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.66: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 31ms, Average = 13ms C:\>ping 172.18.110.66 Pinging 172.18.110.66 with 32 bytes of data: Reply from 172.18.110.66: bytes=32 time=24ms TTL=126 Reply from 172.18.110.66: bytes=32 time=3ms TTL=126 Reply from 172.18.110.66: bytes=32 time=1ms TTL=126 Reply from 172.18.110.66: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.66: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 24ms, Average = 7ms </pre>

REFLECTION

What have you learned in this task?

This assignment taught me how important routing and subnetting are to creating and maintaining an efficient network. Subnetting facilitates effortless interaction between various network segments and helps in the efficient use of IP addresses. Connecting subnets requires proper routing configuration, particularly when utilizing dynamic protocols like RIP. In order to avoid communication problems, I also discovered that changing a router's IP address or subnet requires updating network configurations. Devices on different subnets won't be able to communicate if these updates aren't transmitted over the routing protocol.